

[PHYS 6268] Nonlinear Dynamics: Problem Set 10

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Strogatz Book: Problems 8.4.2, 8.4.12, 8.5.2, 8.6.1, & 8.7.5

1 Problem 8.4.2: Discuss the bifurcations of the system $\dot{r} = r(\mu - \sin r)$, $\dot{\theta} = 1$ as μ varies.

Given the system defined in polar coordinates,

$$\dot{r} = r(\mu - \sin r) \quad ; \quad \dot{\theta} = 1 \quad ,$$

we consider the different topological pictures admitted by the phase portrait for $r > 0$ when μ is varied.

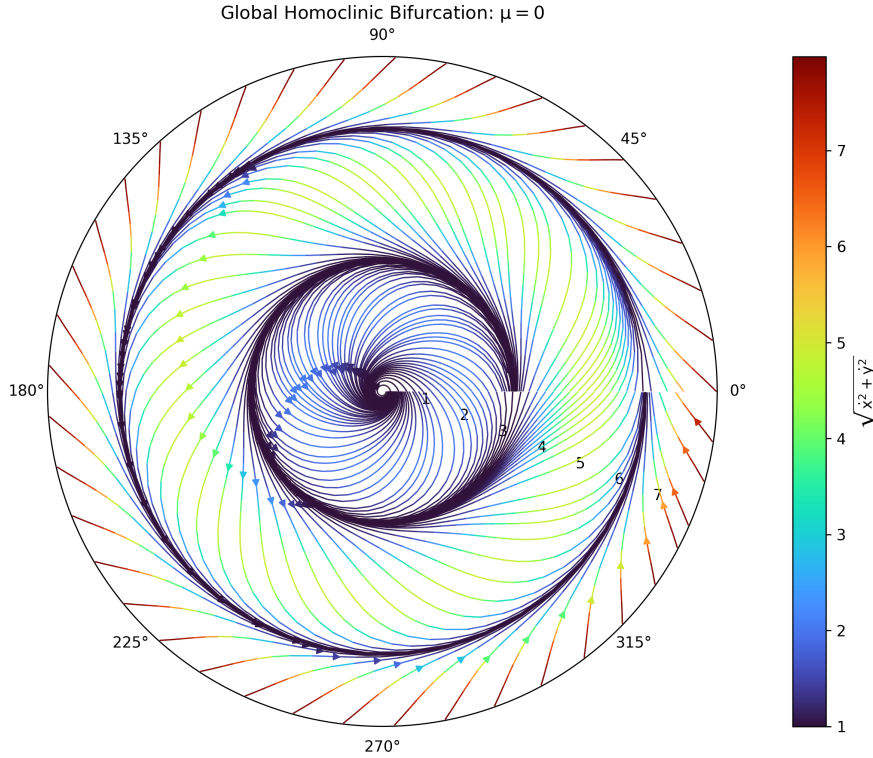


Figure 1: Phase portrait (state space) for the system $\dot{r} = r(\mu - \sin r)$; $\dot{\theta} = 1$, with $\mu = 0$. The magnitude of the gradient corresponds to the perceptually uniform colormap. Note that the lack of continuity of trajectories seen crossing the $\theta = 0$ threshold does *not* correspond to a discontinuous vector field at this value.

From the analytical behavior of the expression for $\dot{r}$, we can tell that a critical behavior will occur as μ crosses the value of $\mu_c = 1$, since at this point the expression in parentheses becomes always positive. Because $r > 0$, for $\mu > \mu_c$ we have that $\dot{r} > 0$ and $\dot{\theta} > 0$, and the equations describe a system that is an unstable spiral out from the origin. In order to investigate all of the different changes that occur as μ is varied, we consider values of $\mu = 0$, $0 < \mu < \mu_c$, $\mu = \mu_c = 1$ and $\mu > \mu_c$. Then, we can build on our understanding for $\mu > 0$ to extrapolate the behavior of the system as μ is *decreased* below zero.

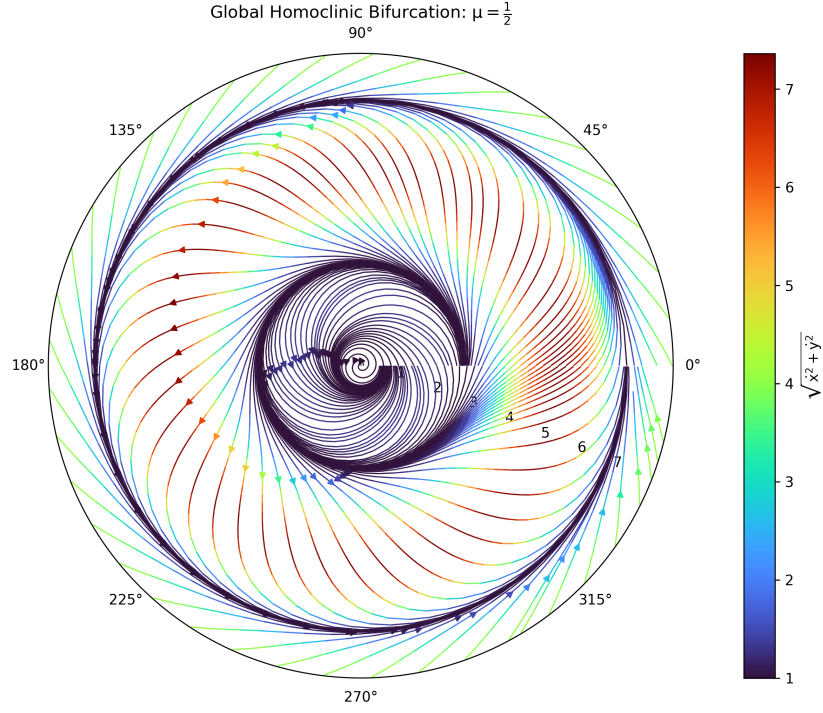


Figure 2: Phase portrait for the system $\dot{r} = r(\mu - \sin r)$; $\dot{\theta} = 1$, with $\mu = 1/2$. The magnitude of the gradient corresponds to the colormap.

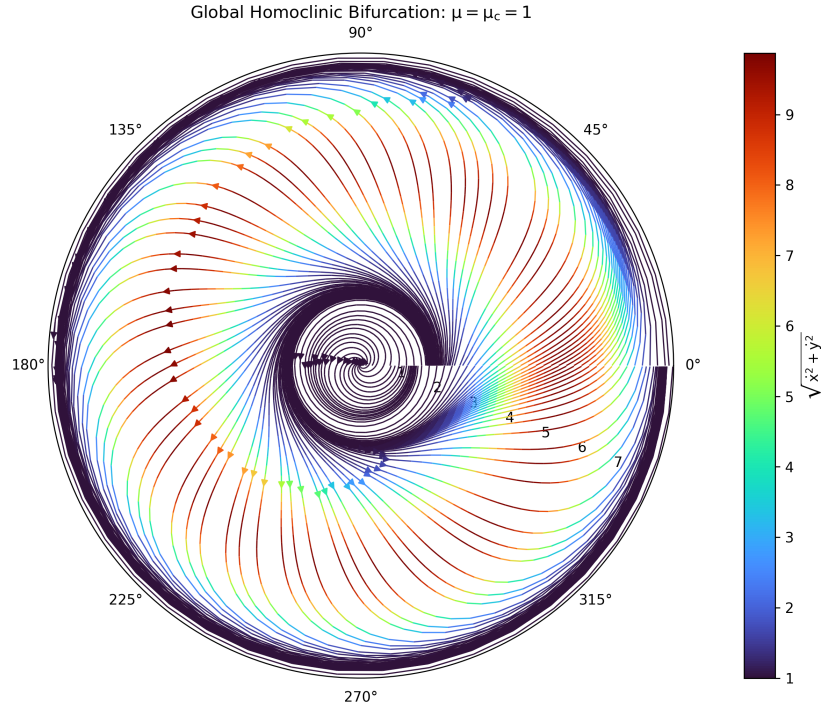


Figure 3: Phase portrait (state space) for the system $\dot{r} = r(\mu - \sin r)$; $\dot{\theta} = 1$, when μ transitions through the critical value $\mu_c = 1$. The critical behavior is visible as the bifurcated stability of the limit cycles surrounding the origin. The magnitude of the gradient corresponds to the perceptually uniform colormap.

Due to the angular symmetry (no θ dependence in $\dot{r}$, $\dot{\theta}$), limit cycles and non-singular fixed points will occur in manifolds that take the shape of a ring or circle around the origin. We begin by considering the system at $\mu = 0$. In this case, $\dot{r} = -r \sin r$. Here, there is a semi-stable fixed point at the origin. In practice, this node acts entirely stably as a spiral inward because $r > 0$ by definition and the stable side is when the origin is approached from $r > 0$. At larger values of r , we encounter a limit cycle for the second intersection of $\dot{r} = 0$. This limit cycle is unstable, as trajectories just inside of it approach the origin, and trajectories just outside approach a *larger radius* limit cycle. This behavior is illustrated in Figure 1, where the magnitude of the gradient is evidently maximized at the points farthest from any limit cycle or fixed point, i.e., directly between them.

The occurrence of a countably infinite number of limit cycles as r increases allows us to analyze only the behavior over a unique “period” and extend the understanding to successive, periodic annuluses. The appearance of periodic limit cycles is straightforward to motivate when considering the equation for $\dot{r}$ in isolation: via the dependence on the $\sin r$ term, each intersection of $\dot{r} = 0$ will have an alternating sign for the derivative $\partial_r \dot{r}$. Since, in the r dimension, the slope of $\dot{r}$ at a fixed point determines the stability, each intersection of $\dot{r} = 0$ corresponds to an oppositely-signed crossing of the r axis, and thus the stability of each successive limit cycle flips.

As μ increases above $\mu = 0$ to a value between zero and unity, the appearance of another intersection with $\dot{r} = 0$ prompts the creation of another limit cycle. Specifically, a stable limit cycle emanates from the origin as μ increases. This limit cycle is attractive both to trajectories just inside and just outside of it: this means that our stable inward spiral is bifurcated into the stable limit cycle just outside of the origin, with an *unstable* fixed point replacing it at the origin. This behavior is evident in the phase portrait depicted by Figure 2, where we fix $\mu = 1/2$. As seen in the Figure, the limit cycles still have an alternating attractiveness / stability as r increases.

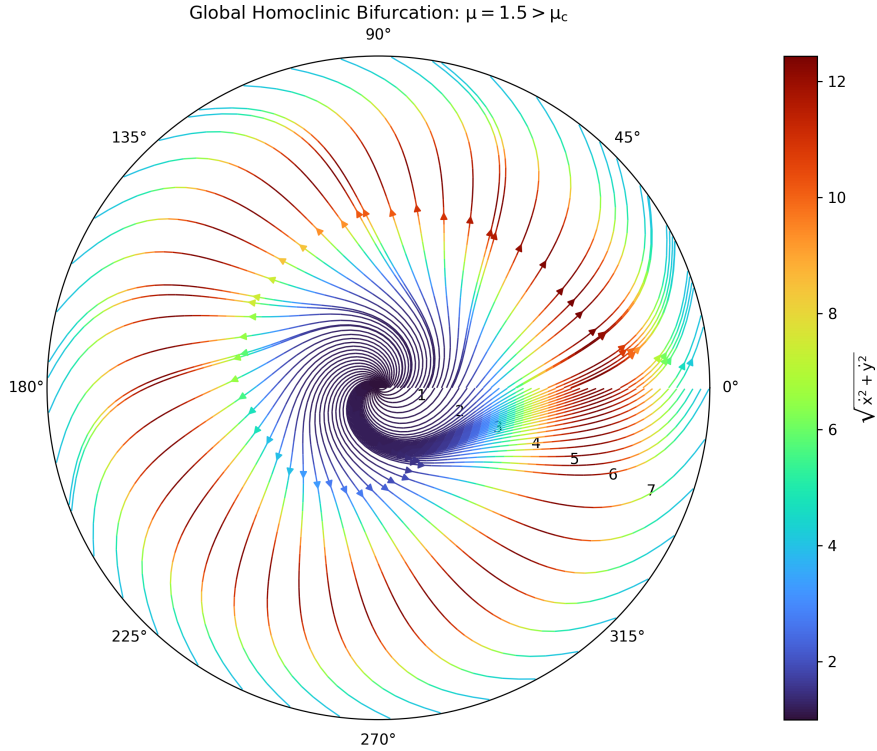


Figure 4: Phase portrait describing the system $\dot{r} = r(\mu - \sin r)$; $\dot{\theta} = 1$, when $\mu = 3/2$.

Next, we consider the system as it approaches criticality: $\mu \rightarrow \mu_c$. This transition is accompanied by the convergence of successive pairs of unstable and stable limit cycles. At this point, only

the “inner” side of each limit cycle is stable to trajectories as time approaches infinity. Trajectories initialized just across the manifold in the radial direction quickly spiral outward towards the next limit cycle. This behavior is identifiable in the phase portrait in Figure 3.

Finally, we consider the case where μ has increased *beyond* the critical regime at $\mu = 1$. We let, e.g., $\mu = 3/2$. Now the system becomes simple. The limit cycles of alternating stability collided at $\mu = \mu_c$ and they are *annihilated* for larger values of μ . This *fold*, or saddle-node bifurcation of limit cycles, has a normal form given in Strogatz. As can be seen in Figure 4, this leaves *only* the unstable spiral at the origin. Hence, trajectories flay outwards radially from the origin monotonically, but as they travel outward they encounter regions of alternating speed. This can be seen by comparing the color (speed) at, e.g., $r = 3$ and $r = 8$ (blue/green) to that at $r = 5$ (orange/red). This pattern continues as r increases.

Note that thus far, we consider only the transition from $\mu = 0 \rightarrow \mu_c \rightarrow \mu > \mu_c$, but not the reverse trend as μ decreases below zero. As μ decreases below zero to negative values, now the behavior for small, positive r is flipped: for $-1 < \mu < 0$, the fixed point at the origin is stable, and an unstable limit cycle exists surrounding it. A stable limit cycle occurs at a larger radius, and another unstable one at a yet larger one. This continues as r increases indefinitely, and the stability of the limit cycles consistently alternate at successively larger values of r that satisfy $\sin r = -1/2$, akin to the behavior we analyze above.

At $\mu = -\mu_c$, we have an interesting picture. The pair of stable/unstable limit cycles coalesce into a single, quasi-stable limit cycle at each 2π -periodic spacing for the intersections of $\dot{r} = 0$. On the “inner” side of each limit cycle manifold, trajectories unstably are attracted away towards $r = 0$. For all limit cycles but the inner-most one, this is another limit cycle that the trajectories tend towards. However, trajectories initialized with coordinates inside of the interior limit cycle will tend towards the origin in a stable inward spiral. As μ decreases below $\mu = -1$, in analogy to the bifurcation at $\mu = \mu_c$, the limit cycles are destroyed at all positive values of r at which they exist. This leaves only the stable spiral inward to the origin remaining, the “mirror” of the system at $\mu = 1$.

2 Problem 8.4.12: Scaling near a homoclinic bifurcation.

To find how the period of a closed orbit scales as a homoclinic bifurcation is approached, we estimate the time it takes for a trajectory to pass by a saddle point (this time is much longer than all others in the problem). Suppose the system is given locally by $\dot{x} \approx \lambda_u x$, $\dot{y} \approx -\lambda_s y$. Let a trajectory pass through the point $(\epsilon, 1)$, where $\epsilon \ll 1$ is the distance from the stable manifold. How long does it take until the trajectory has escaped from the saddle, say out to $x(t) \approx 1$?

Please note that we use ϵ for the dimensionless parameter measuring the initial distance, not μ (as Strogatz uses) to avoid confusion with the system parameter μ from other sections (e.g., see Problem 1). That is, $\mu \equiv \epsilon$ henceforth in Section 2.

Here we are concerned with the timescale T for the trajectory to escape a region on the 2-D unit interval. Near the saddle point equilibrium, we assume (via the prompt) that the system is appropriately represented by the linearization about the equilibrium point. A saddle always has at least a 1-D eigenspace (i.e., at least one eigenvector) that is stable, and at least a 1-D eigenspace that is unstable. Hence, in two dimensions, there is one stable and one unstable dimension. We let x, y here represent the coordinates in phase space such that each dimension bears along one of the two (stable or unstable) manifolds. Via the definition of the system, the y coordinate is stable,

$$\dot{y} = -\lambda_s y \quad ,$$

since it has a negative real part of the eigenvalue. In contrast, the x direction is unstable, following

$$\dot{x} = \lambda_u x \quad .$$

Hence, the motion along the coordinates is entirely *decoupled*, and it is sufficient to consider only the growing (unstable) x dimension when assessing the timescale for the trajectory to “breach” the $x = 1$ surface.

Then, our dynamics in this dimension are described by

$$\frac{dx}{dt} = \lambda_u x \quad .$$

Since we “pin” our trajectory to a specific position in phase space $(\epsilon, 1)$, we immediately have a means to directly integrate the above for a time-dependent solution of $x(t)$ describing the system’s behavior in the neighborhood of the saddle point at the origin. This allows us to write (excusing sloppy integration variable notation):

$$\begin{aligned} \frac{dx}{dt} &= \lambda_u x \\ \implies \frac{dx}{x} &= dt \lambda_u \\ \implies \int_{x(t=0)}^{x(t=t)} \frac{dx}{x} &= \int_{t=0}^{t=t} dt \lambda_u \\ \implies \ln \left| \frac{x(t)}{x(0)} \right| &= (t - 0) \lambda_u \\ \implies x(t) &= x(0) e^{\lambda_u t} \\ x(0) = \epsilon &\implies x(t) = \epsilon e^{\lambda_u t} \quad . \end{aligned}$$

Hence, we have computed an approximate analytical solution for the unstable coordinate of the trajectory through phase space. Now, we evaluate

$$x(t = T) = 1 = \epsilon e^{\lambda_u T} \quad .$$

Taking the natural logarithm and rearranging yields

$$\ln(\epsilon^{-1}) = \lambda_u t \quad ,$$

which can be solved for our critical escape timescale T :

$$T = \frac{\ln\left(\frac{1}{\epsilon}\right)}{\lambda_u} = -\frac{\ln \epsilon}{\lambda_u} \quad .$$

Hence, since $\epsilon \ll 1$, the timescale is positive-definite and it depends on proximity to the stable manifold via

$$T \sim \mathcal{O}(\ln \epsilon) \quad .$$

We also mention that since the stable y coordinate began at $y = 1$, it has steadily *decreased* as the x coordinate increases. Thus, the parametric trajectory through phase space in x, y coordinates curves from one face to an adjacent one. Accounting for their combined motion, this curve takes the shape of an arc or an elbow depending on the magnitudes of the eigenvalues.

3 Problem 8.5.2: Driven Pendulum.

Consider the driven pendulum $\phi'' + \alpha\phi + \sin \phi = I$. By numerical computation of the phase portrait, verify that if α is fixed and sufficiently small, the system’s stable limit cycle is destroyed in a homoclinic bifurcation as I decreases. Show that if α is too large, the bifurcation is an infinite-period bifurcation instead.

To do so, we compare the system as I is increased from $I = 0$ to $I = 2$ for $\alpha \ll 1$ and for $\alpha > 1$. See Figure 5 for the results of this computation. The plot is a function of $\dot{\phi}(\phi)$. The left

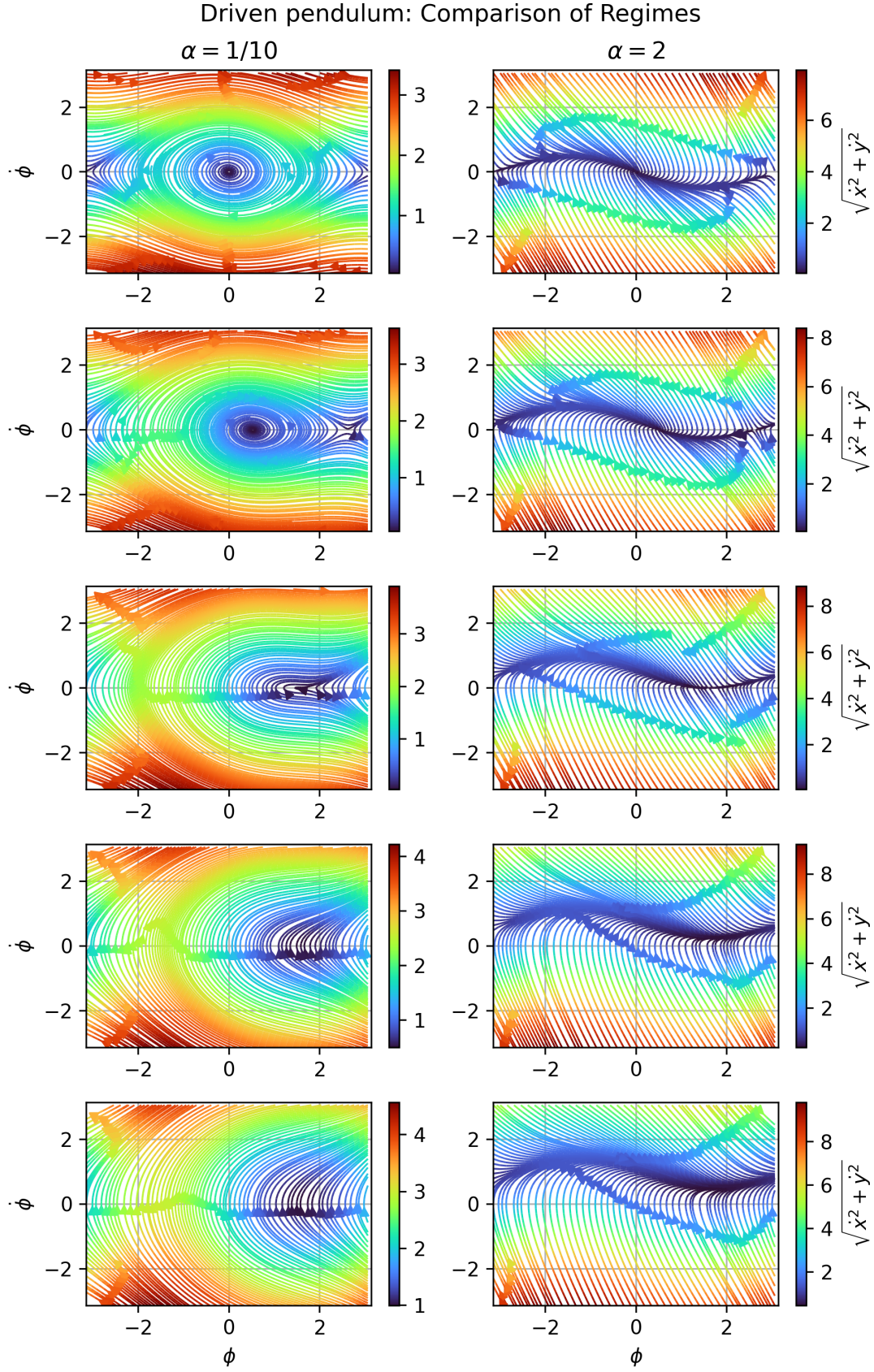


Figure 5: Array of phase portraits for the system of Problem 5: $\ddot{\phi} + \alpha\dot{\phi} + \sin\phi = I$ for α, I being positive parameters. The left column depicts the system for various I (increasing moving downward) and $\alpha = 1/10$. The right column depicts the same system for corresponding I values with $\alpha = 2$. I ranges through values in the set $I \in [0, \frac{1}{2}, 1, \frac{3}{2}, 2]$.

column corresponds to $\alpha = 1/10$, and results are shown for increasing values of I moving downward in the figure. The right column corresponds to $\alpha = 2$, for the same values of I . The magnitude of the first and second derivatives of ϕ are depicted by the colormap. The second order system was rewritten as a pair of coupled first order equations in order to visualize it, with $\dot{\phi}$ serving as the vertical coordinate. The phase space $(\phi, \dot{\phi})$ is topologically similar to a cylinder.

As can be seen via direct comparison, the behavior is qualitatively distinct in the two cases of α regimes. We begin with the case of small α ; in this regime, assuming the other parameters and all derivatives to be $\mathcal{O}(1)$, the system is approximately given by

$$\ddot{\phi} + \sin \phi = I \quad ,$$

which is the equation for a driven harmonic oscillator that experiences no damping. Since I is a constant value, the harmonic oscillator behavior does not engage in any resonance behavior with the driving force. Instead, the solutions represent an analog of simple harmonic motion with scaled amplitudes for small I .

At first, $I = 0$, and the solutions are merely the trajectories described by harmonic motion. That is, they possess stable centers at the origin and homoclinic loops connecting the unstable fixed points at $\phi = \pm\pi$. That behavior is visible in the top panel in the left column of Figure 5 (corresponding to small α). At the next value of $I = 1/2 \neq 0$, a stable limit cycle arises at a nonzero amplitude, shown by the next panel downward in the left column. These solutions represent trajectories that coalesce to a stable limit cycle corresponding to the driving amplitude I ; since $I < 1$, the potential and kinetic energy (position $\sin \phi$ and velocity $\dot{\phi}$) can exchange in the same manner as for an unforced oscillator. However, it is critical for the nature of the bifurcation to note that the fixed points characterized as saddle points *remain* for this configuration. Then, as I grows past unity (i.e., moving down the same column), the stable limit cycle coalesces with these saddle points and a homoclinic bifurcation is encountered. Moving down the rows from the second to the fourth displays this behavior clearly. As I continues to grow, the system exhibits a collection of saddle points with the stable regions of state space gone.

In the alternate regime where $\alpha = 2 > 1$, the topological evolution as I increases behaves in an entirely different way, as shown in the right column of Figure 5. At small I , a stable node exists at the origin. These stable nodes are periodic in space but carry trajectories inward towards the origin directly, rather than being brought into an orbit. This can be rationalized physically as now the damping term *dominates* the system behavior. An overdamped system does not exhibit oscillations, but rather slowly “falls” towards equilibria with no overshoot. An example of this motion is akin to the first problem of the first homework, namely a pendulum moving through a viscous liquid such as honey. As evidenced by the Figure, as I grows larger, the stable contraction along the horizontal ϕ dimension disappears, and the saddle points converge with the stable nodes until a stable manifold exists along a 1D curve. The destruction of the stable orbit as I varies occurs on an unbounded timescale, and thus we recover an infinite-period bifurcation.

4 Problem 8.6.1: “Oscillator Death” and Bifurcations on a Torus.

In a paper on systems of neural oscillators, Ermentrout and Kopell (1990) illustrated the notion of “oscillator death” with the following model:

$$\dot{\theta}_1 = \omega_1 + \sin \theta_1 \cos \theta_2 \quad , \quad \dot{\theta}_2 = \omega_2 + \sin \theta_2 \cos \theta_1 \quad ,$$

where $\omega_1, \omega_2 > 0$.

4.1 (a) Sketch all the qualitatively different phase portraits that arise as ω_1, ω_2 vary.

To analyze changes to the qualitative topological nature of the phase portraits for this system as ω_1, ω_2 vary, we first consider the governing equations. The system is symmetric, i.e., replacing

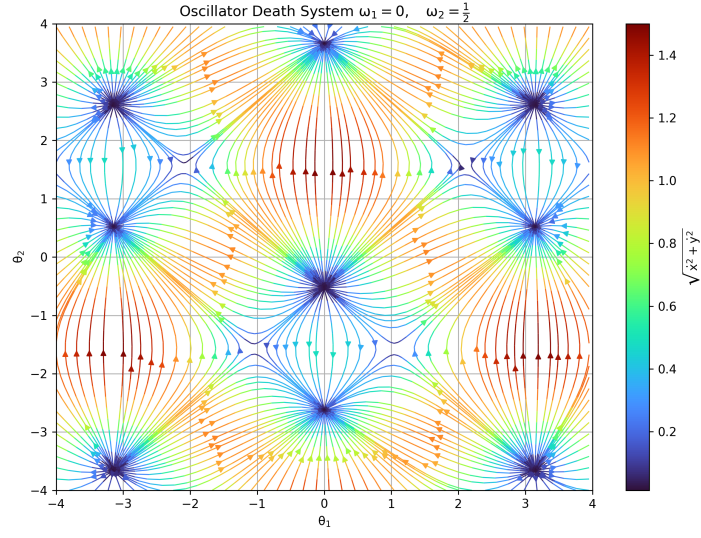


Figure 6: System in Problem 4 with $\omega_1 = 0, \omega_2 = 1/2$.

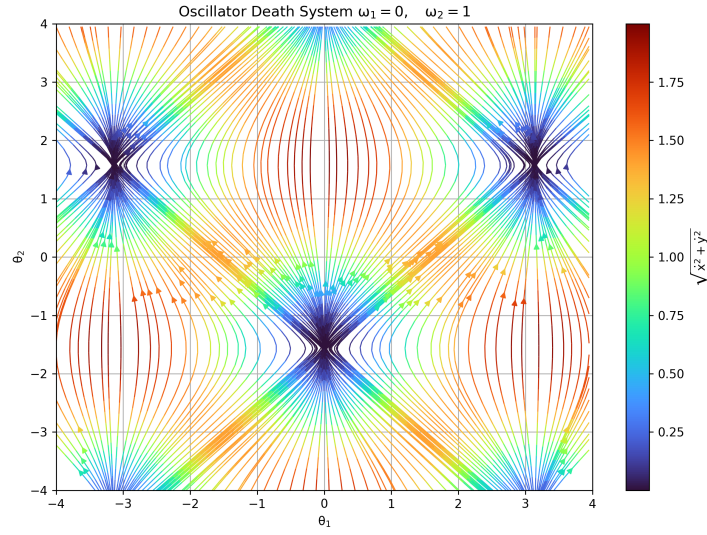


Figure 7: System in Problem 4 with $\omega_1 = 0, \omega_2 = 1$. The value of $\omega_2 = \omega_c = 1$ represents the critical transition of the system in the θ_2 (vertical) direction.

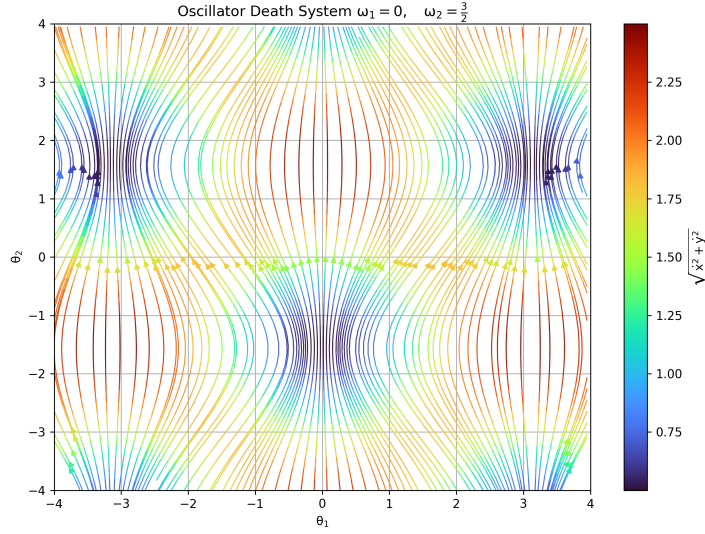


Figure 8: System in Problem 4 with $\omega_1 = 0, \omega_2 = 3/2$. This represents the system in a supercritical state.

$(\theta_1, \omega_1) \mapsto (\theta_2, \omega_2)$ only reorders the equations (but preserves their mathematical structure). Then, we consider just one of the equations to start:

$$\dot{\theta}_1 = \omega_1 + \sin \theta_1 \cos \theta_2 \quad .$$

This expression has substantially different behavior for three regimes: $\omega_1 = 0$, $\omega_1 < 1$, $\omega_1 \geq 1$.

Since we are told that $\omega_1 > 0$, this leaves only the two regimes across the critical value $\omega_c = 1$. The topological structure of the arrangement of trajectories in phase space depends mostly on when the expression for $\dot{\theta} = 0$. Since the image of an arbitrary product of trigonometric functions is bounded on the unit interval, we know that *in the “worst case”*, $\sin \theta_1 \cos \theta_2 = 1$. Then, $\omega_1 \geq 1$ means that $\dot{\theta}_1 > 0$ for all time: hence, the system has equilibria (stable and unstable) for $\omega_1 < 1$, but not above this value.

Therefore, we depict the phase portrait for three separate conditions. We fix $\omega_1 = 0$ in order to illustrate the transition through increasing values of $\omega_2 = 1/2$, $\omega_2 = 1$, and $\omega_2 = 3/2$. Then, we extend this to when both parameters are varied by showing the corresponding cases when both frequencies $\omega_1 = \omega_2 \in [1/2, 1, 3/2]$.

Figure 6 depicts the system for the combination $(\omega_1, \omega_2) = (0, \frac{1}{2})$. In this case, it is evident that the system has a periodic lattice (extending in both θ_1 and θ_2) of fixed points occurring in regular groups. The stability of each fixed point alternates in a diametrically adjacent fashion. The stable and unstable fixed points are connected between by saddle points. As ω_2 is increased to unity, the critical transition occurs in the vertical dimension: as shown in Figure 7, the fixed points coalesce into a lattice of saddle points. Since ω_1 remains below the critical value (note that for both $\omega_1, \omega_2 \neq 0$ the critical value is a bit more complex, see Section 4.2), the horizontal coordinate still exhibits periodic stable behavior. The trajectories “fan out” in θ_1 and reconvene back at the next successive saddle point, all flowing “upward” in the θ_2 direction.

As ω_2 grows above unity, these saddle points are destroyed and there are no remaining equilibria. As can be seen in Figure 8, the flow along θ_2 is so dominant that it carries the trajectories smoothly upward before they can coalesce into a saddle that is attractive along θ_1 . Hence, the oscillatory behavior of the system is killed by this bifurcation.

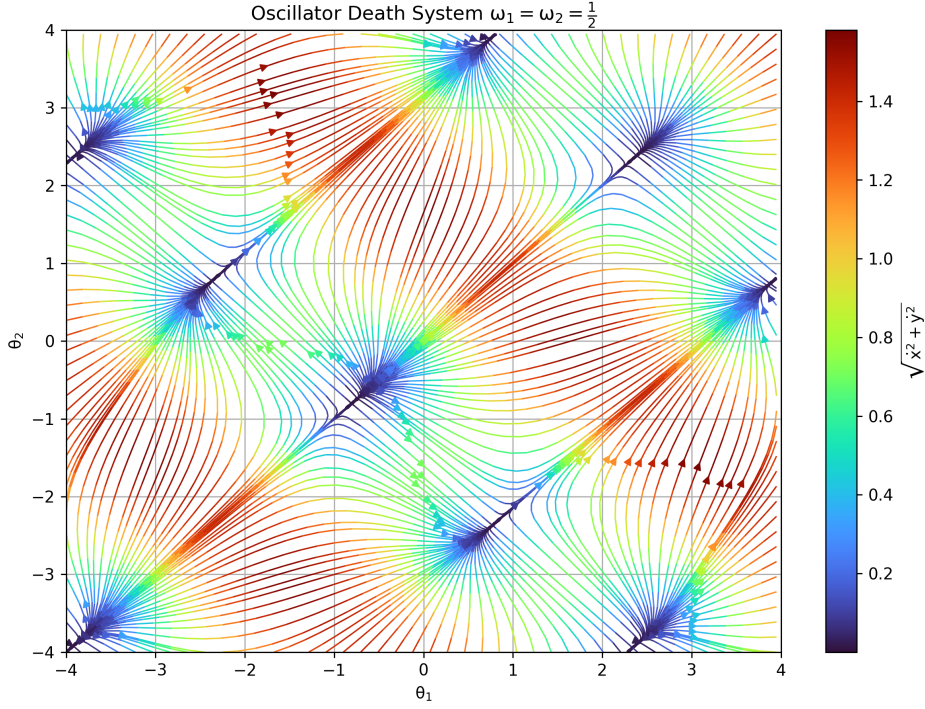


Figure 9: System in Problem 4 with $\omega_1 = 1/2$, $\omega_2 = 1/2$. Hence, the values of ω_1, ω_2 are riding on the curve separating stable / unstable fixed points from only saddles (i.e., at the threshold of “oscillator death”).

4.2 (b) Find the curves in ω_1, ω_2 space along which bifurcations occur, and classify the various bifurcations.

To do so, we first notice that replacing $\Theta \equiv \theta_1 + \theta_2$ yields

$$\dot{\Theta} = \omega_1 + \omega_2 + \sin \Theta \quad . \quad (1)$$

This immediately yields us a means to calculate the bifurcation curve in ω_1, ω_2 space. Originally, we would have to deal with the deeply coupled nature of the two simultaneous equations. Now, we consider only the Θ coordinate now, which increases along the outward positive diagonal emanating from the origin. Then,

$$\dot{\Theta} = 0 \implies \omega_1 + \omega_2 = -\sin \Theta \quad ,$$

which we can see only has a solution when $\omega_1 + \omega_2 \leq 1$. Hence, $\omega_1 + \omega_2 = 1$ is the curve along which the bifurcations to our system occur. This is consistent with our treatment in Section 4.1: since $\omega_1 = 0$ for those Figures, $\omega_2 = 1$ satisfies the above condition. We can check this further by allowing $\omega_1 = \omega_2 = 1/2$: this case is depicted in Figure 9. Akin to the notion of $\theta_1 + \theta_2$ being the relevant coordinate and with equal values of $\omega_1 = \omega_2$, the trajectories “flow” diagonally upward and to the right.

4.3 (c) Plot the stability diagram in ω_1, ω_2 parameter space.

The stability diagram is included in Figure ?? . The lower region corresponds to stable and unstable nodes separated by saddles, and the upper region corresponds to oscillating speed but monotonic “runaway” of the trajectories. There are no stable regions in phase space above the

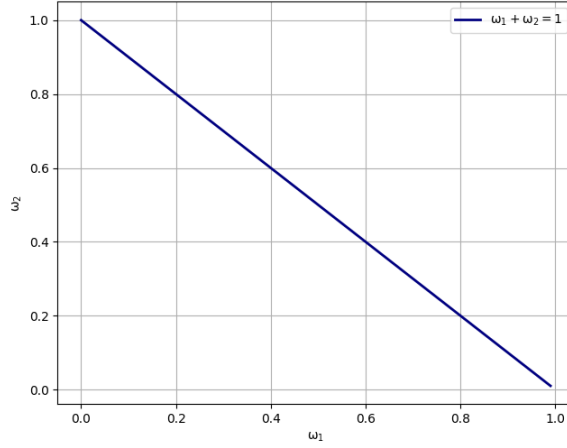


Figure 10: Stability diagram for the coupled oscillator system in Section 4.

curve. On the curve, the system exhibits only saddle connections with a stable dimension largely along the coordinate coupled to the subcritical (i.e., the smaller) ω .

5 Problem 8.7.5: Another Driven Overdamped System.

By considering an appropriate Poincaré map, prove that the system $\dot{\theta} + \sin \theta = \sin t$ has at least two periodic solutions. Can you say anything about their stability?

To do so, we recast the system as a vector field over a cylindrical domain: this relies on treating the temporal coordinate t as a variable. Hence, $\dot{t} = 1$, that is, time is constantly progressing. This corresponds to the axial dimension of the cylinder in phase space. Then, the θ behavior follows

$$\dot{\theta} = \sin t - \sin \theta \quad .$$

Since t has a constant progression, only the θ coordinate admits nullclines. These take the shape of

$$\sin t = \sin \theta \quad .$$

Since $\sin t \in [-1, 1]$, this always admits solutions (at all times, that is) for certain values of θ .

We wish to utilize the nullclines to infer an appropriate Poincaré map. The map must be oriented transverse to the flow. Since t “flows” along the axial direction of the cylinder regardless of the behavior in θ , we know the map construction must have a component of its surface normal vector along the axial dimension of our cylindrical phase space geometry. However, since we want the trajectory to intersect this surface multiple times, it will be mostly oriented perpendicular to the “winding” motion around the cylinder, and the axial component of the surface normal will be small.

The nullclines suggest that when $\dot{\theta} = 0$, $\theta = 2\pi n - \arcsin \sin t$ for some integer n . Our Poincaré mapping using the surface described above is of the form

$$P(\theta) : \mathbb{R} \mapsto \mathbb{R} \quad ,$$

calculated using the time of flight for a periodic trajectory.

6 Software Usage

The entirety of the work was carried out on pen and paper prior to typesetting, except for the plotting: this was completed using python (`numpy` and `matplotlib`).